Database Environment

CSE 341
Introduction to Database Systems
Winter 2024
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Introduction

- Main objective of a database is to provide users with an abstract view of data, hiding certain details of how data is stored and manipulated.

- The starting point for the design of a database must be an abstract and general description of the information requirements of the organization.

- To accommodate these needs and more, the architecture of most commercial DBMSs available today is based to some extent on the so-called ANSI-SPARC architecture (a three-level architecture).

Objectives

- Three-level ANSI-SPARC architecture
- Types of language that are used by DBMSs,
- The concepts of data models and conceptual modeling
- Functions that we would expect a DBMS to provide.

Objectives of a Three Level Architecture

- All users should be able to access same data.

- A user's view* is immune to changes made in other views.

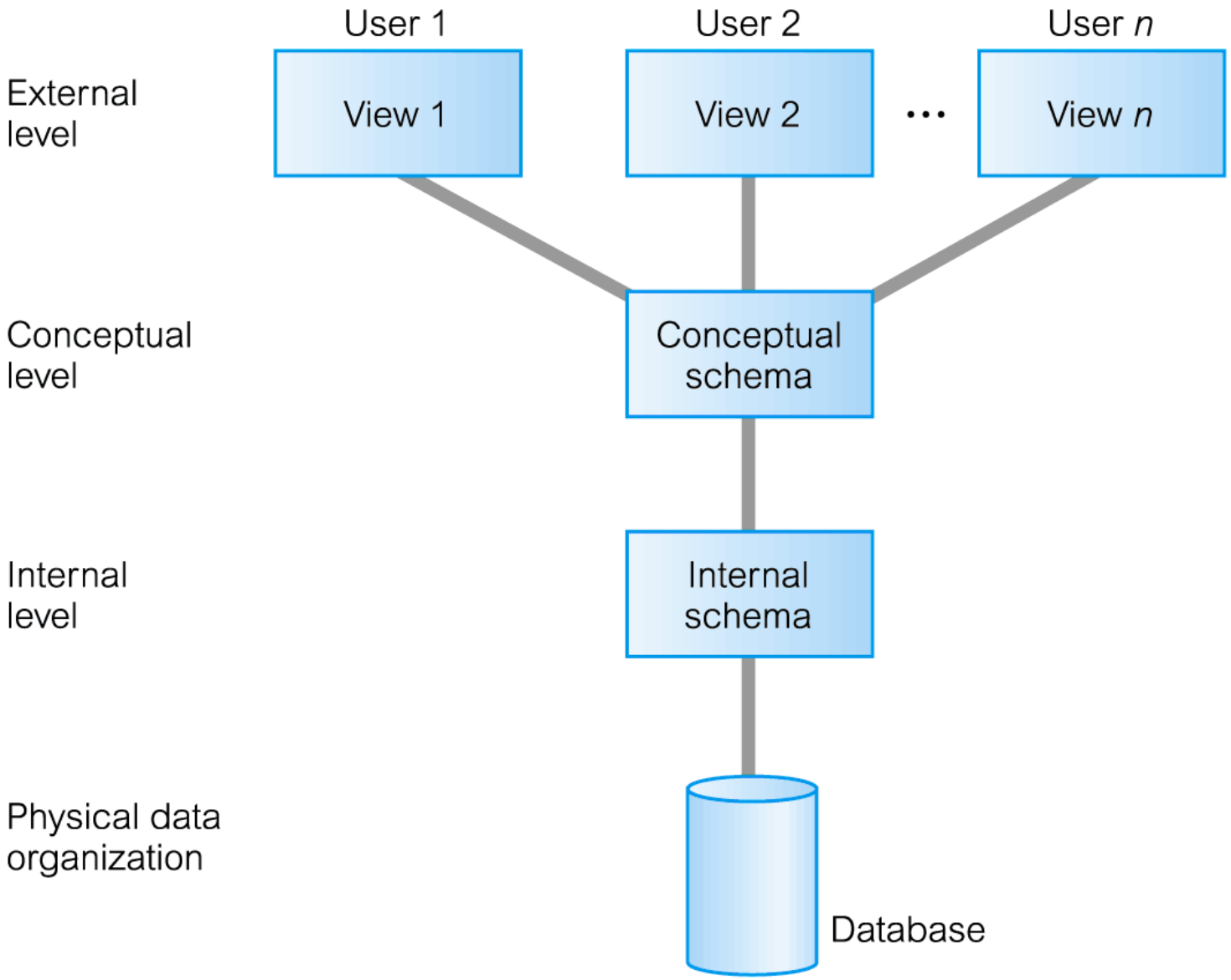
  **each user may require a different view of the data held in the database.*

- Users should not need to know physical database storage details.

Objectives of a Three Level Architecture

- DBA should be able to change database storage structures without affecting the users' views.
- Internal structure of database should be unaffected by changes to physical aspects of storage.
- DBA should be able to change conceptual structure of database without affecting all users.

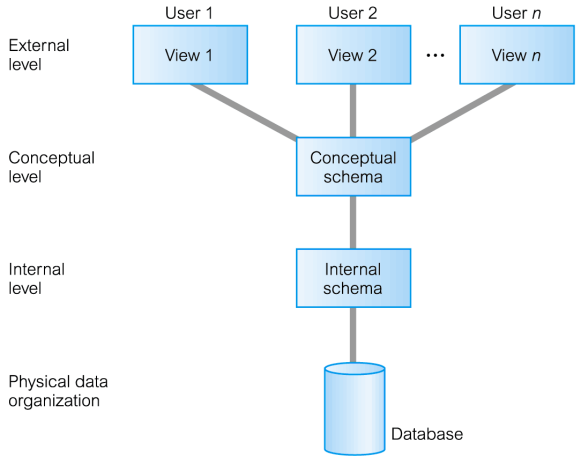
ANSI-SPARC Three-Level Architecture



ANSI-SPARC Three-Level Architecture

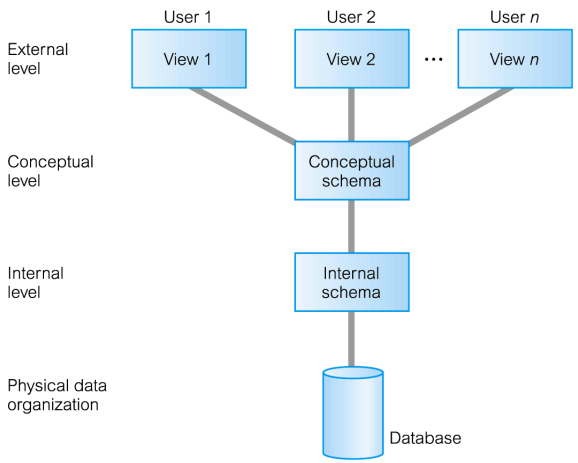
- External Level
  - Users' view of the database.
  - Describes that part of database that is relevant to a particular user.

- Conceptual Level
  - Community view of the database.
  - Describes what data is stored in database and relationships among the data.



ANSI-SPARC Three-Level Architecture

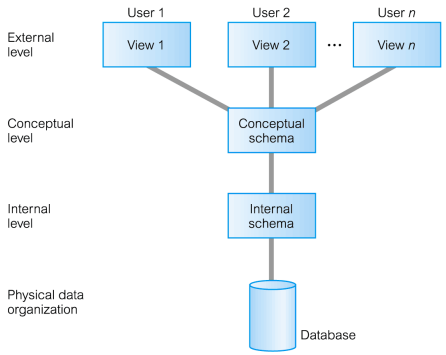
- Internal Level
  - Physical representation of the database on the computer.
  - Describes how the data is stored in the database.



Schemas

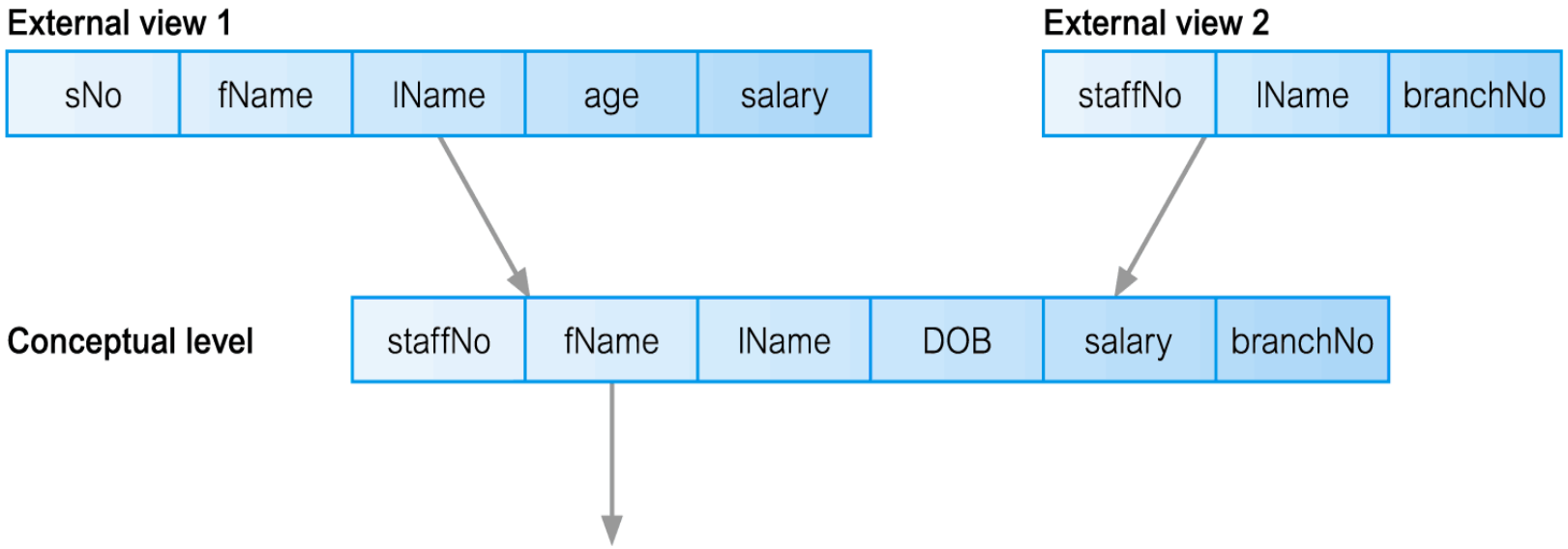
- The overall description of the database is called the database schema.
- There are three different types of schema in the database and these are defined according to the levels of abstraction of the three-level architecture

Schemas



- At the highest level, we have multiple external schemas (also called subschemas) that correspond to different views of the data.
- At the conceptual level, we have the conceptual schema, which describes all the entities, attributes, and relationships together with integrity constraints.
- At the lowest level, we have the internal schema, a complete description of the internal model, containing the definitions of stored records, the methods of representation, the data fields, and the indexes and storage structures used.
- There is only one conceptual schema and one internal schema per database.

Differences between Three Levels of ANSI-SPARC Architecture



Internal level

```
struct STAFF {
    int staffNo;
    int branchNo;
    char fName [15];
    char lName [15];
    struct date dateOfBirth;
    float salary;
    struct STAFF *next;                /* pointer to next Staff record */
};
index staffNo; index branchNo;         /* define indexes for staff */
```

Data Independence

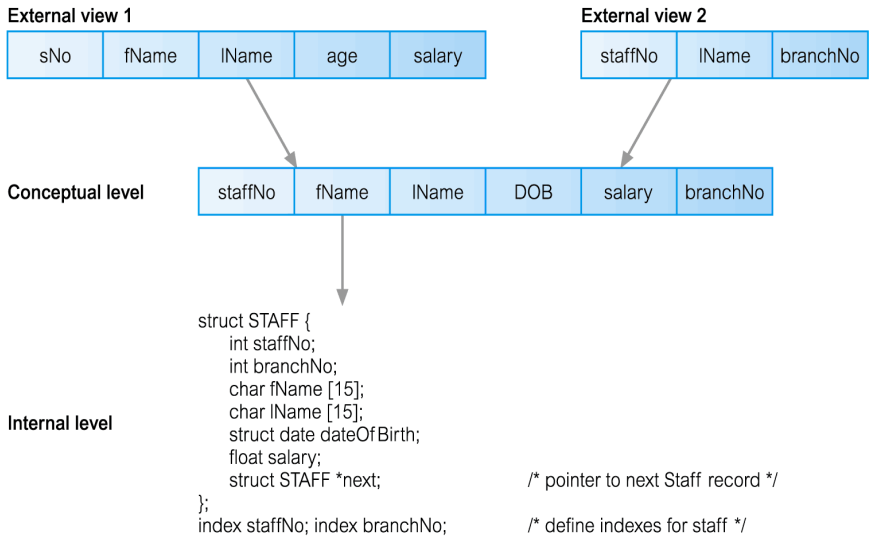
- A major objective for the three-level architecture is to provide data independence,
  - upper levels are unaffected by changes to lower levels.

- There are two kinds of data independence: <u>logical</u> and <u>physical</u>.

Data Independence

Logical Data Independence

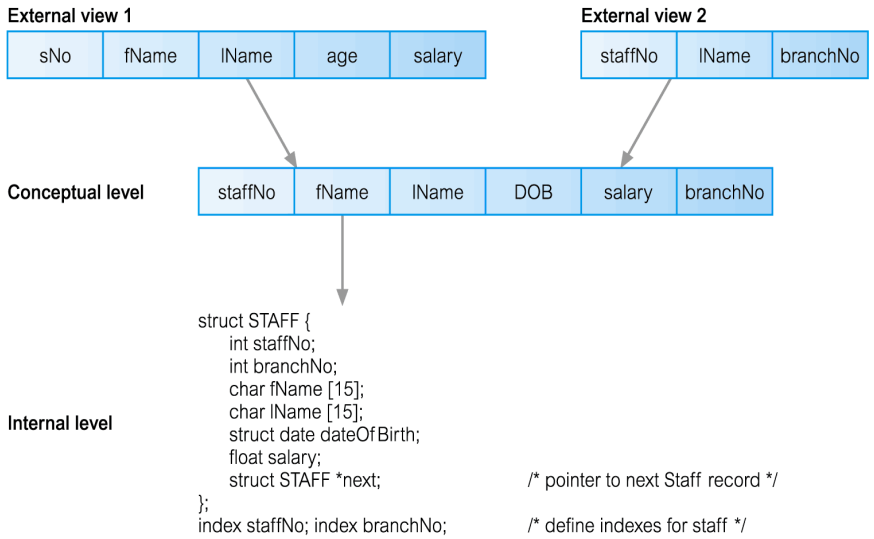
- Refers to immunity of external schemas to changes in conceptual schema.
- Conceptual schema changes (e.g. addition/removal of entities).
- Should not require changes to external schema or rewrites of application programs.



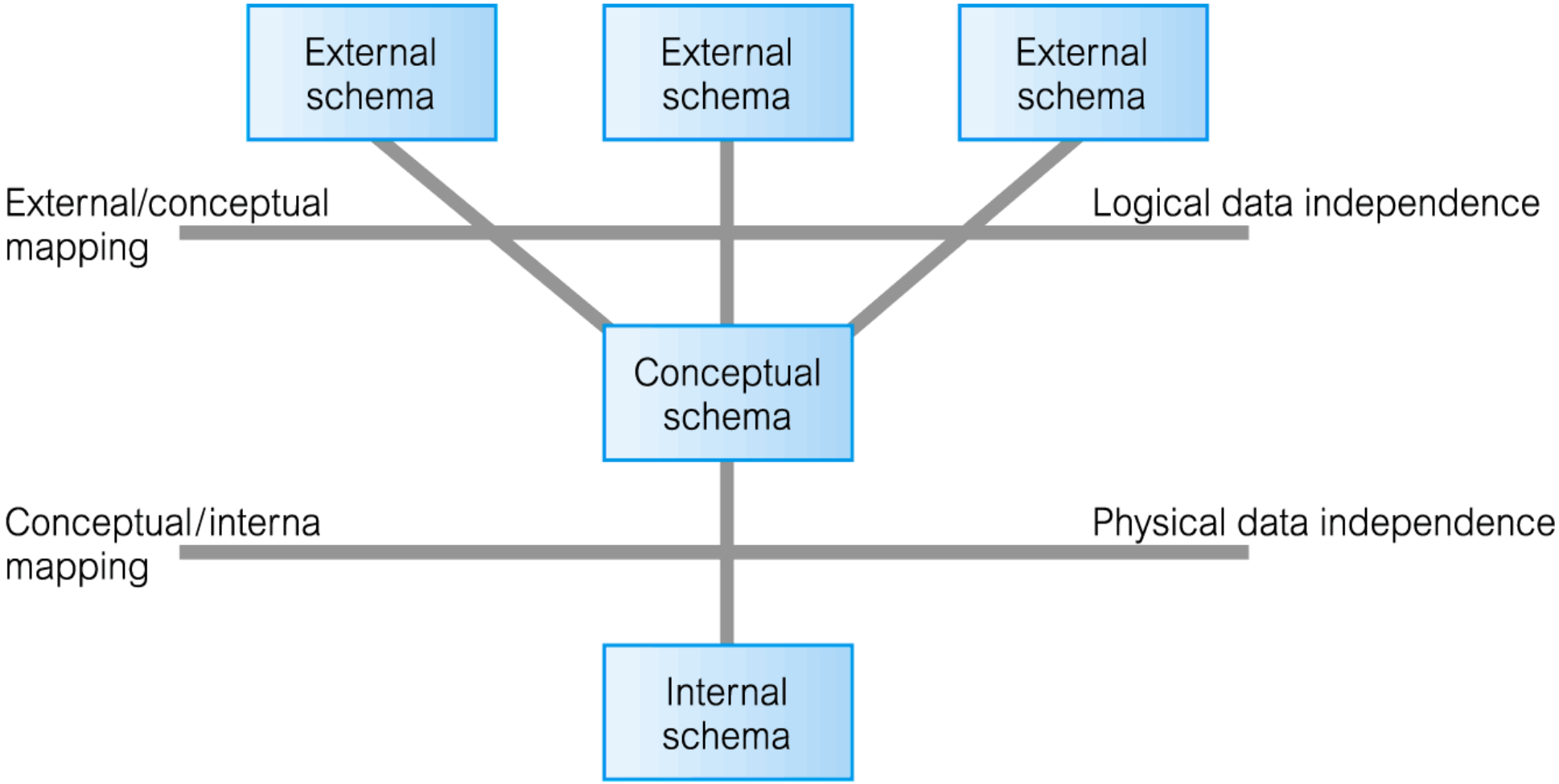
Data Independence

Physical Data Independence

- Refers to immunity of conceptual schema to changes in the internal schema.
- Internal schema changes (e.g. using different file organizations, storage structures/devices).
- Should not require change to conceptual or external schemas.



Data Independence and the ANSI-SPARC Three-Level Architecture



Database Languages

Data Definition Language (DDL)

- Allows the DBA or user to describe and name entities, attributes, and relationships required for the application
- + any associated integrity and security constraints.

Database Languages

Data Manipulation Language (DML)

- Provides basic data manipulation operations on data held in the database.

Procedural DML

- allows user to tell the system exactly how to manipulate data.

Non-Procedural DML

- allows user to state what data is needed rather than how it is to be retrieved.

Fourth Generation Languages (4GLs)

- a shorthand programming language. The user defines *what* is to be done, not *how* (e.g. report generators, SQL)

Data Model

We mentioned that a schema is written using a DDL.

- Too low level to describe the data requirements of an organization.
- Need for a higher-level description of the scheme: i.e. a data model

Integrated collection of concepts for describing data, relationships between data, and constraints on the data in an organization.

Data Model comprises:

- a structural part;
- a manipulative part;
- possibly a set of integrity rules.

Data Model

- Purpose
  - To represent data in an understandable way.

- Categories of data models include:
  - Object-based
  - Record-based
  - Physical.

Data Models

- Object-Based Data Models
  - Entity-Relationship
  - Object-Oriented.
- Record-Based Data Models
  - Relational Data Model
  - Network Data Model
  - Hierarchical Data Model.
- Physical Data Models

Relational Data Model

Branch

branchNo	street	city	postCode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Staff

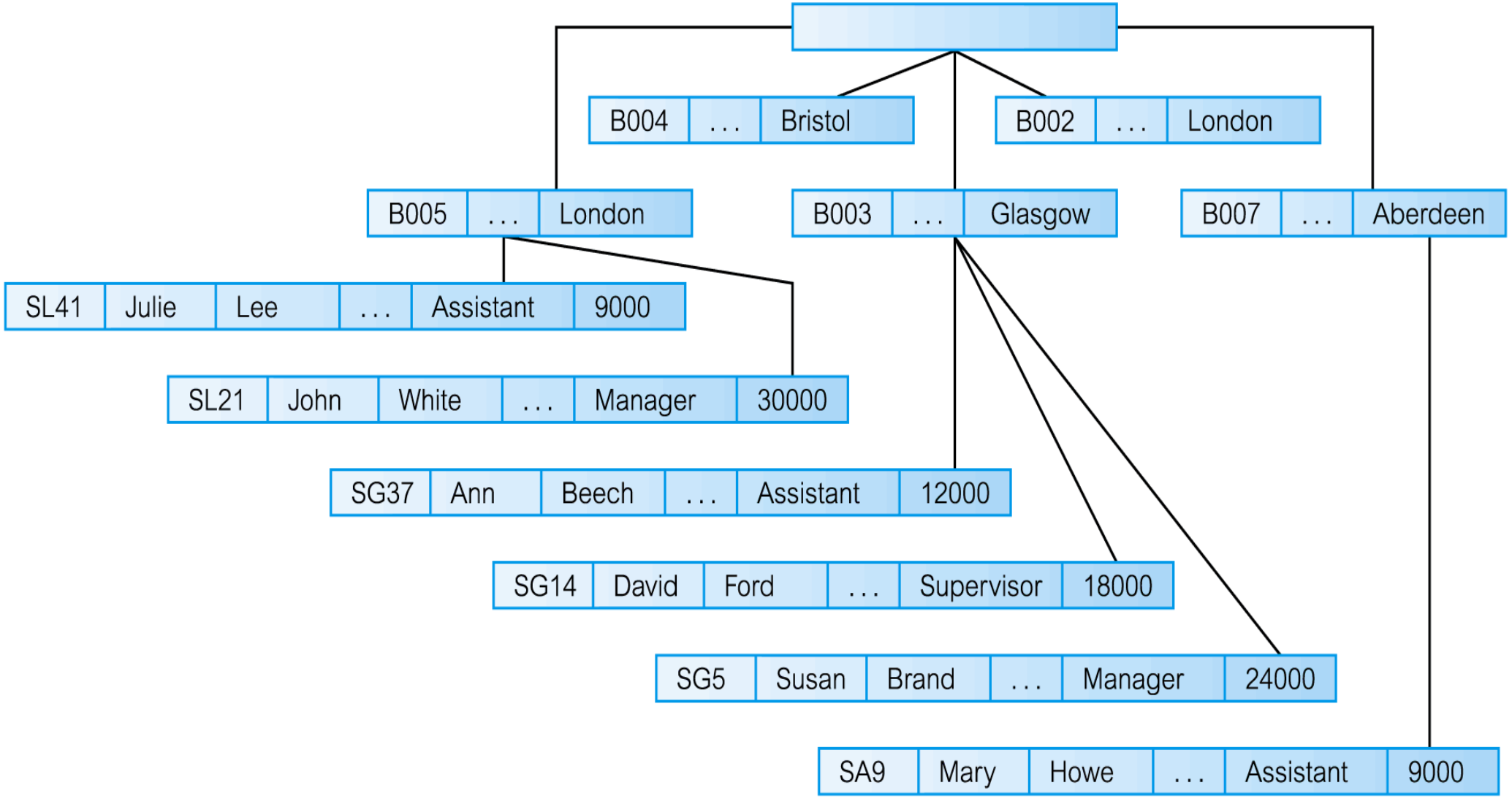
staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Network Data Model

B005	22 Deer Rd	London
B007	16 Argyl St	Aberdeen
B003	163 Main St	Glasgow
B004	32 Manse Rd	Bristol
B002	56 Clover Dr	London

SL41	Jul e	Lee	. . .	Assistant	9000
SL21	John	White	. . .	Manager	30000
SA9	Mary	Howe	. . .	Assistant	9000
SG37	Ann	Beech	. . .	Assistant	12000
SG14	David	Ford	. . .	Supervisor	18000
SG5	Susan	Brand	. . .	Manager	24000

Hierarchical Data Model



Conceptual Modeling

- Conceptual schema is the core of a system supporting all user views.
- Should be complete and accurate representation of an organization's data requirements.

- Conceptual modeling is process of developing a model of information use that is independent of implementation details.
- Result is a conceptual data model.

Functions of a DBMS

- Data Storage, Retrieval, and Update.
- A User-Accessible Catalog.
- Transaction Support.
- Concurrency Control Services.
- Recovery Services.

Functions of a DBMS

- Authorization Services.
- Support for Data Communication.
- Integrity Services.
- Services to Promote Data Independence.
- Utility Services.

System Catalog: Data about the Data

- Repository of information (metadata) describing the data in the database.
- One of the fundamental components of DBMS.
- Typically stores:
  - names, types, and sizes of data items;
  - constraints on the data;
  - names of authorized users;
  - data items accessible by a user and the type of access;
  - usage statistics.

Review Questions

- Explain the concept of database schema and discuss the three types of schema in a database.
- What is a data model? Discuss the main types of data model.
- Discuss the function and importance of conceptual modeling.
- Discuss the differences between DDL and DML. What operations would you typically expect to be available in each language?

Next…

Database Architectures and the Web